

ECO 310 Midterm Study Priorities

Final Push for 2026 Exam

Updated: March 4, 2026

Current Exam Readiness: 75–80%

Excellent Progress!

You've made major improvements since your last review. Many high-priority concepts have moved to green, including critical topics like substitution effects, perfect substitutes, and Hicksian demand.

1 Concepts You've Mastered

Recent Wins (Previously Red, Now Green)

- **Perfect substitutes bang-per-buck** (sp25_mc5) — Was RED!
- **Substitution effect decomposition** (sp25_mc6) — Was RED!
- **Hicksian demand with expenditure** (sp25_sa3c) — Was RED!
- **Hicksian with perfect complements** (sp25_mc3)
- **Certainty equivalent** (sp24_sa2a)
- **Risk aversion definition** (sp24_mc6)
- **WARP** (sp24_mc9)
- **Corner solutions & Cobb-Douglas** (sp24_mc2, mc3)
- **Portfolio optimization** (sp24_sa2bb)

Core Strengths

- **Cobb-Douglas spending rule** — All questions green
- **Risk aversion applications** — Multiple questions mastered
- **Writing EU expressions** — sp23_sa4a, sp24_sa2ba
- **Basic FOC optimization** — sp23_sa4bi
- **WARP basics** — sp23_mc2, sp24_mc4
- **Perfect substitutes** — sp24_mc10, sp25_mc5

2 Critical Gaps — Urgent Attention Needed

High Priority for 2026 Exam

1. **sp24_mc5: Income effect with perfect complements**

Why critical: Review session emphasized perfect complements heavily

Key concept: With $u(x_1, x_2) = \min\{x_1, x_2\}$, substitution effect = 0, so ALL change is income effect.

Action: Review 2024 MC5 solution completely

2. **sp25_mc4: Portfolio risk/return calculation**

Why critical: Asset allocation appears in review session

Key formulas:

- Portfolio $\sigma = \alpha \times \text{risky asset } \sigma$
- Expected return = $\alpha r_{\text{risky}} + (1 - \alpha)r_{rf}$
- Remember: variance $\neq$ standard deviation!

3. **sp24_sa3d: Hicksian demand (intertemporal)**

Why critical: Expenditure minimization was in 2026 review Q1

Action: Similar to your now-green sp25_sa3c, but with time twist

Lower Priority Red (Less Likely on Exam)

- **sp23_mc5:** Independence axiom violation (appeared in 2026 review, possible MC)
- **sp23_sa2aii:** EU proofs with weighted sums (complex, lower probability)
- **sp24_mc1:** Rationality/transitivity (lexicographic example in review)
- **sp23_sa3a, sp23_sa3d:** Non-standard utility (only 2023, less likely)
- **sp24_sa3c:** Marshallian intertemporal (light in recent years)

3 Yellow Status — Quick Review Needed

Most Important

1. **sp25_sa1a, 1b, 1c: CES utility (all three parts)**

Quick review of limits:

- $s \rightarrow 0$: perfect complements
- $s \rightarrow 1$: Cobb-Douglas
- $s \rightarrow \infty$: perfect substitutes

2. **sp25_sa3bii: Corner solution explanation**

Related to $p \geq 1/2$ derivation for interior solutions

3. **sp25_sa2b: Log utility wealth threshold**

Appeared in both 2024 & 2025

4. **sp23_sa4bii: Corner solutions in EU**

Related to portfolio problems

4 Today's Study Plan (Next 2–3 Hours)

Session 1: Perfect Complements Income Effect (30 min)

Review 2024 MC5:

- Original: $L = R = 5$ at $(p_L = 1, p_R = 1)$, income $m = 10$
- New: p_R rises to 4
- New optimal: $L = R = \frac{10}{1+4} = 2$
- **Substitution effect = 0** (no reoptimization with $\min\{\}$)
- **Income effect = total change = 2 - 5 = -3**

Session 2: Portfolio Calculations (30 min)

Key concepts:

- Variance vs. standard deviation: $\sigma = \sqrt{variance}$
- Portfolio std dev: $\sigma_p = \alpha \cdot \sigma_{risky}$
- Portfolio expected return: $E[r_p] = \alpha E[r_{risky}] + (1 - \alpha)r_{rf}$

Practice: 2025 MC4

Session 3: Hicksian with Time (45 min)

Work through 2024 SA3d:

- Similar structure to 2026 review Q1
- Minimize cost subject to utility constraint
- Compare to your now-green sp25_sa3c

Session 4: Quick CES Review (30 min)

Review the three limit cases and practice identifying which form applies

5 2026 Exam Predictions

Based on the official review session:

Almost Guaranteed Topics

- **Perfect complements:** Marshallian, Hicksian, and expenditure function
- **Compensating variation:** Possibly with perfect substitutes
- **Expected utility:** Multi-state problem with probability trees
- **Certainty equivalent & risk premium:** Using correct definition $u(w_0 + CE) = EU$
- **FOC for portfolio/betting:** With corner solution analysis
- **WARP:** With numerical price/bundle data

Very Likely

- Cobb-Douglas spending rule (MC)
- Perfect substitutes corner solutions (MC)
- Substitution vs. income effects
- Independence axiom or continuity (MC)

6 Key Formulas & Concepts

Perfect Complements: $u(x_1, x_2) = \min\{\alpha x_1, \beta x_2\}$

Optimal spending: $\alpha x_1 = \beta x_2$

Marshallian: $x_1 = \frac{\beta m}{\alpha p_1 + \beta p_2}, x_2 = \frac{\alpha m}{\alpha p_1 + \beta p_2}$

Hicksian: $x_1^c = \frac{\beta u^*}{\alpha}, x_2^c = \frac{\alpha u^*}{\beta}$

Expenditure: $e(p_1, p_2, u^*) = \left(\frac{\beta p_1}{\alpha} + p_2\right) \cdot \frac{\alpha u^*}{\beta}$

Key insight: Substitution effect = 0 (no reoptimization possible)

Cobb-Douglas: $u(x_1, x_2) = x_1^\alpha x_2^\beta$

Magic rule: Spend $\frac{\alpha}{\alpha + \beta}$ of income on good 1

Perfect Substitutes: $u(x_1, x_2) = \alpha x_1 + \beta x_2$

Bang-per-buck: Buy only good 1 if $\frac{\alpha}{p_1} > \frac{\beta}{p_2}$

Risk Aversion

Definition: $u(E[W]) > EU$ (utility of expected wealth beats expected utility)

Implies: $u''(w) \leq 0$ (concave utility)

Certainty equivalent: $u(w_0 + CE) = EU$

Risk premium: $\pi = E[W] - CE$

Portfolio Optimization

EU maximization: Choose x to maximize $EU(x)$

Interior solution: $EU'(x) = 0$

Corner solution: If $EU'(0) < 0$, invest nothing; if $EU'(x_{\max}) > 0$, invest maximum

Final Checklist Before Exam

Master sp24_mc5 (income effect with perfect complements)

Master sp25_mc4 (portfolio variance/std dev)

Review sp24_sa3d (Hicksian intertemporal)

Quick CES limits review (sp25_sa1b)

Verify corner solution logic (sp25_sa3bii)

Review 2026 lecture Q1 (perfect complements all three demands)

Review 2026 lecture Q2 (CV with perfect substitutes)

Review Eagles/Chiefs betting example (FOC & interpretation)

You're 75–80% ready!

Focus on the critical red topics and you'll be fully prepared.
Your recent progress has been excellent — keep up the momentum!